

Venture Capital and Entrepreneurial Management

MGMT 264

Valuation Problem Set

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Valuation Problem Set - Q1a-b

Items	Venture Capitalist	
Investment	\$5M	
Investment Horizon	5 years	
Comparable Firms' P/E	20X	
Projected Net Income	\$5M in year 5	
Required Rate of Return	50%	30%
Initial Capital Structure	1M shares outstanding	
Required % Ownership	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$	$\$5M / [(20 * \$5M) / 1.3^5] \approx 18.6\%$
# of New Shares	$1M * 38\% / (1 - 38\%) = 612,903$	$1M * 18.6\% / (1 - 18.6\%) = 228,501$
Price/Share	$\$5M / 612,903 = \8.16	$\$5M / 228,501 = \21.88
Post-Money Valuation	$\$5M / 38\% = \$13.17M$	$\$5M / 18.6\% = \$26.88M$
Pre-Money Valuation	$\$13.17M - \$5M = \$8.17M$	$\$26.88M - \$5M = \$21.88M$

Formula derivation: $38\% = X / (1000000 + X)$ implies $x = 1M * (38\% / (1 - 38\%))$

Valuation Problem Set - Q1c

Items	Venture Capitalist	
Investment	\$12M	
Investment Horizon	5 years	
Comparable Firms' P/E	20X	
Projected Net Income	\$5M in year 5	
Required Rate of Return	50%	30%
Initial Capital Structure	1M shares outstanding	
Required % Ownership	$\$12M / [(20 * \$5M) / 1.5^5] \approx 91.1\%$	$\$12M / [(20 * \$5M) / 1.3^5] \approx 44.6\%$
# of New Shares	$1M * 91.1\% / (1 - 91.1\%) = 10.23M$	$1M * 44.6\% / (1 - 44.6\%) = 803,594$
Price/Share	$\$12M / 10.23M = \1.17	$\$12M / 803,594 = \14.93
When Investment=\$5M	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$	$\$5M / [(20 * \$5M) / 1.3^5] \approx 18.6\%$

■ Main results:

- Roger must give away much more of the company in return for the higher investment sought
- Post-money valuation does not change, because post-money (in this case where there is no dilution!) is simply the discounted terminal value of the company *) ($\$12M / .911 = \$13.17M$)
- However, the pre-money valuation changes ($\$13.17M - \$12M = \$1.17M$)

Valuation Problem Set - Q2

Items	Venture Capitalist	
Investment	\$5M	
Investment Horizon	5 years	
Terminal Value	\$100M	
Required Rate of Return	50%	30%
Initial Capital Structure	1M shares outstanding, 15% management reserve	
Final % Ownership	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$	$\$5M / [(20 * \$5M) / 1.3^5] \approx 18.6\%$
Retention Ratio	$1 - 15\% = 85\%$	$1 - 15\% = 85\%$
Current % Ownership	$38\% / 85\% = 44.7\%$	$18.6\% / 85\% = 21.88\%$

- **Note:** Implicit assumption is that the VC will require that the 15% management pool be set up at the time of financing.

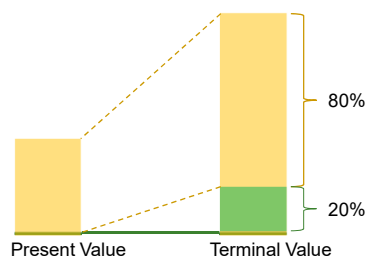
Valuation Methodologies

- An Illustration of Dilution Formulas

Formula 1: Retention ratio =

100% - cumulative % of future dilution

If the company issues new shares to new investors (or management) which enable them to **hold 20% at liquidity**

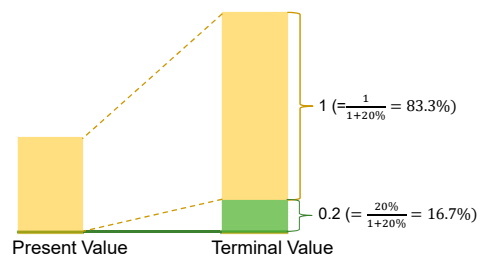


$$\text{Retention Ratio} = 1 - 20\% = 80\%$$

Formula 2: Retention ratio =

$1/(1 + \text{\% of dilution in round T})/\dots$

If the company issues **additional 20%** new shares to new investors (or management) **before liquidity**



$$\text{Retention Ratio} = \frac{1}{1+20\%} = 83.3\%$$

Valuation Problem Set - Q2

Items	Venture Capitalist	
Investment	\$5M	
Investment Horizon	5 years	
Terminal Value	\$100M	
Required Rate of Return	50%	30%
Initial Capital Structure	1M shares outstanding, 15% management reserve	
Final % Ownership	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$	$\$5M / [(20 * \$5M) / 1.3^5] \approx 18.6\%$
Retention Ratio	$1 / (1 + 15\%) = 87\%$	$1 / (1 + 15\%) = 87\%$
Current % Ownership	$38\% / 87\% = 44\%$	$18.6\% / 87\% = 21\%$

■ **Note:** Implicit assumption is that the company will issue additional 15% shares to management after the financing.

Valuation Problem Set - Q2

Items	Venture Capitalist	
Required Rate of Return	50%	
Future Dilution	Yes	No
Final % Ownership	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$
Current % Ownership	$38\% / 85\% = 44.7\%$	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$
# of New Shares	$1M * 44.7\% / (1 - 44.7\%) = 808,318$	$1M * 38\% / (1 - 38\%) = 612,903$
Price/Share	$\$5M / 808,318 = \6.19	$\$5M / 612,903 = \8.16
Post-Money Valuation	$\$5M / 44.7\% = \$11.19M$	$\$5M / 38\% = \$13.17M$
Pre-Money Valuation	$\$11.19M - \$5M = \$6.19M$	$\$13.17M - \$5M = \$8.17M$

■ **Consequently:**

- More shares will have to be issued (since higher current ownership acquired)
- Share price will be lower (same investment for more shares)
- Post-money valuation will be lower, pre-money as well, because now % ownership acquired > final % ownership, and post-money = investment / % ownership acquired

Valuation Problem Set - Q3a-b

	Investor 1	Investor 2
Terminal Value	\$100M	\$100M
IRR	50%	30%
Investment Horizon	5 years	3 years
Investment	\$5M	\$3M
Initial shares	1M shares outstanding, 15% management share reserve	
Final % Ownership	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$	$\$3M / [(20 * \$5M) / 1.3^3] \approx 6.6\%$
Retention Ratio	$1 - 15\% - 6.6\% = 78.4\%$	$1 - 15\% = 85\%$
Current % Ownership	$38\% / 78.4\% = 48.46\%$	$6.6\% / 85\% = 7.8\%$
# of New Shares	$[48.46\% / (1 - 48.46\%)] * 1M = 940,241$	$[7.8\% / (1 - 7.8\%)] * (1M + 0.940241M) = 163,337$
Price/share	$\$5M / 940,241 = \5.32	$\$3M / 163,337 = \18.37
Post-Money Valuation	$\$5M / 48.46\% = \$10.32M$	$\$3M / 7.8\% = \$38.63M$
Pre-Money Valuation	$\$10.32M - \$5M = \$5.32M$	$\$38.63M - \$3M = \$35.63M$

Valuation Problem Set - Q3c

	Investor 1	Investor 2
Terminal Value	\$100M	\$100M
IRR	50%	30%
Investment Horizon	5 years	5 years
Investment	\$5M	\$3M
Initial shares	1M shares outstanding, 15% management share reserve	
Final % Ownership	$\$5M / [(20 * \$5M) / 1.5^5] \approx 38\%$	$\$3M / [(20 * \$5M) / 1.3^5] \approx 11.1\%$
Retention Ratio	$1 - 15\% - \underline{6.6\%} = 78.4\%$	$1 - 15\% = 85\%$
Current % Ownership	$38\% / 78.4\% = 48.46\%$	$11.1\% / 85\% = 13.1\%$
# of New Shares	$[48.46\% / (1 - 48.46\%)] * 1M = 940,241$	$[13.1\% / (1 - 13.1\%)] * (1M + 0.940241M) = 292,487$
Price/share	$\$5M / 940,241 = \5.32	$\$3M / 292,487 = \10.26
Post-Money Valuation	$\$5M / 48.46\% = \$10.32M$	$\$3M / 13.1\% = \$22.9M$
Pre-Money Valuation	$\$10.32M - \$5M = \$5.32M$	$\$22.9M - \$3M = \$19.9M$

Valuation Problem Set - Q3c

■ Pro-rata shares

Items	Investor 1
Ownership Before 2 nd Round	48.46%
Total Shares	940,241
# of Shares in 2 nd Round	292,487
Total Shares after 2 nd Round	$1M + 940,241 + 292,487 = 2,232,728$
New Shares to Buy in Round 2	$(48.46\% * 2,232,728) - 940,241 = \mathbf{141,739}$
Price of Round 2	\$10.26/share
Investment in Round 2	$141,739 * \$10.26/\text{share} = \mathbf{\$1,453,798}$

■ **Note:** This assumes that all round 2 investors have same r.o.r. of 30%!

Valuation Problem Set - Q3c

Time	0	1	2	3	4	5	6	7	IRR
Pro-rata	-\$5M	0	-\$1.453M	0	0	0	0	\$41.191M	32.2%
Diluted	-\$5M	0	0	0	0	0	0	\$35.81M	32.48%
Horizon Change	-\$5M	0	0	0	0	0	0	\$38M	33.6%
No Change	-\$5M	0	0	0	0	\$38M	0	0	50%

■ **Note:**

- In the pro-rata situation, the de facto value to the investor is calculated as $48.46\% \times 0.85 \times \$100M = \$41.191M$. In the dilution case, the de facto value to the investor is calculated as $48.46\% \times (1 - 11.1\% - 15\%) \times \$100M = \$35.81M$.
- The first round investor's IRR is affected most by the extension of the investment horizon from 5 to 7 years, not so much by the additional dilution in year 3.
- If first round VC did **not** suffer from additional dilution in year 3 but in year 7 would have a final ownership of 38% - the one required to deliver a 50% IRR over a five year horizon – then the IRR after 7 years would be 33.6%.

Valuation Problem Set – Q4a

	Investor 1	Founder
Firm's Terminal Value	\$100M	
Required % Ownership	38%	62%
Repaid Principal	\$5M	\$0M
Remaining Value	$\$100M - \$5M = \$95M$	
Participation Value	$38\% \times \$95M = \$36.07M$	$62\% \times \$95M = \$58.93M$
Final Payoff	$\$5M + \$36.07M = \$41.07M$	\$58.93M
De Facto Claim	41.07%	58.93%

Time	0	1	2	3	4	5	IRR
Investor 1	-\$5M	0	0	0	0	\$41.07M	52%

Valuation Problem Set – Q4b

Logic:

- Roger should make a counter offer which gives Benedicta just enough of the company to make a 50% IRR:
- $\$5M + x\% \bullet \$95M = \$38M$ (from Q1)
- $x = 34.7\%$

	Investor 1	Founder
Firm's Terminal Value	\$100M	
Required % Ownership	34.7%	65.3%
Repaid Principal	\$5M	\$0M
Remaining Value	$\$100M - \$5M = \$95M$	
Participation Value	$34.7\% \times \$95M = \$33M$	$65.3\% \times \$95M = \$62M$
Final Payoff	$\$5M + \$33M = \$38M$	\$62M
De Facto Claim	38%	62%

Valuation Problem Set – Q4c

Required % Ownership of Investor: 38%		
Company Value	To Investor	To Founder
\$4M	100%	0
\$5M	100%	0
\$10M	69%	31%
\$20M	54%	47%
\$30M	48%	52%
\$40M	46%	54%
\$50M	44%	56%
\$100M	41%	59%
\$150M	40%	60%

Note:

- Assumption is that investor requires 38% of residual firm value, i.e., 38% of enterprise value less principal of \$5M.
- When company value equal to \$10M, the de facto ownership of investor is calculated as $(\$5M + 38\% \times \$5M) / \$10M = 69\%$; If \$20M, $(\$5M + 38\% \times \$15M) / \$20M = 54\%$; If \$30M, $(\$5M + 38\% \times \$25M) / \$30M = 48\%$; If 100M, $(\$5M + 38\% \times \$95M) / \$100M = 41\%$.

Valuation Problem Set – Q5

■ Dividends Conversion (Not Compounded)

Items	Venture Capitalist
Initial Conditions	Cumulative non-cash dividends of 10% of the original cost
Result from Q1 (IRR=50%)	(1) Price/share=\$8.16; (2) Ownership=38%; (3) # of shares=612,903
Cumulative Dividends	\$5M (Investment) * 10% * 5years=\$2.5M
Dividends Conversion	\$2.5M/\$8.16=306,372 shares
Total shares	612,903 + 306,373=919,275
Final Ownership	919,275/(1M+919,275)=47.8%

Time	0	1	2	3	4	5	IRR
Investor 1	-\$5M	0	0	0	0	\$47.8M	57.1%

- **Note:** Cumulative dividends serve to increase Benedicta's de facto claim on the value created from the nominal 38% equity stake to 47.8%, while increasing her rate of return from 50% to 57.1%.

Valuation Problem Set – Q5

■ Dividends Conversion (Compounded)

Items	Venture Capitalist
Initial Conditions	Cumulative non-cash dividends of 10% of the original cost
Result from Q1 (IRR=50%)	(1) Price/share=\$8.16; (2) Ownership=38%; (3) # of shares=612,903
Cumulative Dividends	\$5M (Investment) * (1+10%) ^{5 years} - \$5M = \$3.05M
Dividends Conversion	\$3.05M/\$8.16=374,087 shares
Total shares	612,903 + 374,087=986,990
Final Ownership	986,990/(1M+986,990)=49.7%

Time	0	1	2	3	4	5	IRR
Investor 1	-\$5M	0	0	0	0	\$49.7M	58.3%

- **Note:** Cumulative dividends serve to increase Benedicta's de facto claim on the value created from the nominal 38% equity stake to 49.7% assuming the dividends compounded, while increasing her rate of return from 50% to 58.3%. **Typically, the dividends in such deals do not compound.**

Your Questions?

